

Gisma
University
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Sciences

Advanced ML Project Challenges

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Project choices 01–20

Images

01 Corruption robustness

02 Few flower labels

03 Pet segmentation

04 Unseen defects

05 Satellite bands

06 Plant domain shift

Time series

07 Several forecast horizons

08 Sparse retail demand

09 Remaining engine life

10 Timely anomaly alarms

11 New sensor users

12 Irregular observations

13 Forecast uncertainty

Text

14 Metadata shortcuts

15 Long review context

16 Active label selection

17 Explanation quality

18 Intent abstention

19 Multiple emotions

20 Multilingual budgets

50 project choices in total. Select one core project per team.



Thirty additional choices

Images

21 Hierarchical labels
22 Noisy training labels
23 Unlabelled image data
24 Knowledge distillation
25 Rare traffic signs
26 Novel characters
27 Adversarial pixels
28 Image denoising
29 Texture representations
30 Continual class learning

Time series

31 Spoken keywords
32 Environmental sounds
33 ECG leads and diagnoses
34 Sleep-stage context
35 Sensor augmentation
36 Appliance disaggregation
37 Long-gap imputation
38 Traffic concept drift
39 Variable lengths
40 Early sequence decisions

Text

41 Spam decision costs
42 Sentence paraphrases
43 Similarity as a score
44 Question word order
45 Entities in noisy text
46 Grammar and rare words
47 Inference shortcuts
48 Evidence retrieval
49 Topic discovery
50 Ordered review ratings

Projects 21–50 extend the same notebook requirements and assessment rubric.

Framework choices

Tools	Role in the projects
scikit-learn	Baselines, preprocessing, validation and metrics
XGBoost / LightGBM / CatBoost	Boosted trees on structured features or lag summaries
PyTorch or Keras	CNNs, LSTMs and autoencoders

Choose one neural-network stack and report its compute budget.

scikit-learn XGBoost LightGBM CatBoost PyTorch examples Keras examples

Official framework repositories. Source/paper repositories on later slides may use older environments.



01 Image corruption robustness

Question	**How much accuracy survives noise, blur and compression?**
Input	CIFAR-10 images and labels, plus CIFAR-10-C corrupted test images.
Task	Compare a CNN with HOG + XGBoost. Test an augmentation ablation across corruption severities.
Deliver	Class probabilities, severity curves, calibration results and annotated failure images.

Data: CIFAR-10 + CIFAR-10-C GitHub: hendrycks/robustness Paper: Hendrycks (2019)

Starter scope: 10,000 training images and 5 declared corruption types. GPU helpful.



02 Learning with few flower labels

Question	**Can a model learn 102 flower categories with very few labels?**
Input	Oxford Flowers 102 images, class labels and official train/validation/test IDs.
Task	Compare frozen CNN features + XGBoost with CNN fine-tuning at 1, 5 and 10 labels per class.
Deliver	Learning curves over 3 label-sampling seeds, macro F1 and a table of confused species.

Data: Oxford Flowers 102 GitHub: pytorch/examples Paper: Nilsback (2008)

Starter scope: official 102-class split, cached image features. GPU helpful for fine-tuning.



03 Pet segmentation and boundaries

Question	**Can a CNN locate an animal accurately at its boundaries?**
Input	Oxford-IIIT Pet photographs and pixel-level trimap masks.
Task	Train a U-Net-like CNN. Compare loss functions and examine errors around thin structures and fur.
Deliver	Predicted masks, overlays, mean IoU, boundary analysis and a loss-function ablation.

Data: Oxford-IIIT Pet GitHub: keras-team/keras-io Paper: Parkhi (2012)

Starter scope: 128 × 128 images and a declared subset. GPU recommended.



04 Defects learned from normal images

Question	**Can an autoencoder find defects it never saw during training?**
Input	One MVTec AD category: normal training images, test defects and reference masks.
Task	Train a convolutional autoencoder on normal images. Compare PCA reconstruction and an optional PatchCore model.
Deliver	Anomaly scores, defect heatmaps, image/pixel AUROC and a documented alarm threshold.

Data: MVTec AD GitHub: Anomalib Paper: Bergmann (2019)

Starter scope: one category at 128 × 128. GPU helpful. Dataset uses a noncommercial licence.

05 RGB versus multispectral learning

Question	**Which extra satellite bands improve land-use classification?**
Input	EuroSAT 13-band Sentinel-2 patches and land-use labels. Derive RGB from the same patches.
Task	Compare XGBoost on spectral features with RGB and 13-channel CNNs on identical image IDs.
Deliver	Band-ablation results, macro F1, confusion matrices and an accuracy/runtime comparison.

Data: EuroSAT GitHub: phelber/EuroSAT Paper: Helber (2019)

Starter scope: a stratified 5,000-image subset. GPU helpful. TIFF preprocessing required.



06 Plant disease across datasets

Question	**Does a model trained on clean leaf photographs work in the field?**
Input	PlantVillage and PlantDoc photographs. Build an explicit map of shared species/disease labels.
Task	Train a CNN on PlantVillage. Compare a frozen-feature XGBoost model and a controlled augmentation.
Deliver	Within-source and cross-dataset macro F1, a label-mapping table and a domain-error analysis.

Data: PlantVillage + PlantDoc GitHub: PlantDoc-Dataset Paper: Singh (2020)

Starter scope: 3–5 verified shared classes. GPU helpful. Substantial data cleaning.



07 Forecasting at several horizons

Question	**When do lagged boosted trees compete with sequence models?**
Input	ETTh1 hourly oil temperature and load measurements from the ETT repository.
Task	Forecast 24 and 96 hours ahead with lag-based XGBoost, an LSTM and a seasonal-naive baseline.
Deliver	Forecast paths, errors by horizon, rolling-origin results and a context-window ablation.

Data: ETTh1 CSV GitHub: zhouhaoyi/ETDataset Paper: Zhou (2021)

Starter scope: ETTh1 and 2 forecast horizons. CPU baseline feasible. GPU helpful for LSTM.



08 Sparse retail demand and hierarchy

Question	**Can forecasts remain accurate and consistent when many sales are zero?**
Input	M5 daily item sales, calendar and prices. Select one store and one product category.
Task	Compare LightGBM and CatBoost with seasonal naive. Aggregate item forecasts into category totals.
Deliver	28-day forecasts, item/aggregate errors, hierarchy consistency and zero-demand diagnostics.

Data: M5 data and guide GitHub: M5-methods Paper: Makridakis (2022)

Starter scope: one store/category subset and 3 rolling origins. CPU feasible. Multi-table joins.

09 Remaining useful engine life

Question	**How many operating cycles remain before an unseen engine fails?**
Input	NASA C-MAPSS FD001 sensor histories, engine IDs and endpoint RUL reference values.
Task	Compare an LSTM with XGBoost on window statistics. Separate engines during validation.
Deliver	RUL for each test engine, RMSE, late-warning errors and representative prediction paths.

Data: NASA C-MAPSS GitHub: pytorch/examples Paper: Saxena (2008)

Starter scope: FD001 and a 30-cycle window. CPU baseline feasible. GPU helpful for LSTM.



10 Timely streaming anomaly alarms

Question	**Can an anomaly detector raise useful alarms early without flooding users?**
Input	Numenta Anomaly Benchmark CSV streams and labelled anomaly windows.
Task	Compare an LSTM or convolutional autoencoder with rolling statistical and Isolation Forest baselines.
Deliver	A causal alarm timeline, detection delay, false alarms and event-level precision/recall.

Data: NAB streams and labels GitHub: numenta/NAB Paper: Lavin (2015)

Starter scope: 3–5 declared streams. CPU baseline feasible. Small autoencoder optional on CPU.



11 Activity recognition for new users

Question	**Does a sensor model recognize activities for people absent from training?**
Input	UCI HAR accelerometer/gyroscope windows, activity labels and subject IDs.
Task	Compare a 1D CNN or LSTM with XGBoost on signal summaries. Hold out whole participants.
Deliver	Activity predictions, subject-level macro F1 and errors under sensor/channel ablations.

Data: UCI HAR GitHub: timeseriesAI/tsai Paper: Anguita (2013)

Starter scope: official 6-activity dataset. CPU baseline feasible. GPU helpful for CNN/LSTM.



12 Irregular and missing data

Question	**Does the pattern of missing measurements help predict the outcome?**
Input	PhysioNet 2012 Set A: first 48-hour observations and its separate outcome file.
Task	Compare LSTM inputs with/without masks and time gaps against XGBoost on patient summaries.
Deliver	Patient risk probabilities, AUROC/AUPRC, calibration and a missingness ablation.

Data: PhysioNet Challenge 2012 GitHub: zhiyongc/GRU-D Paper: Che (2018)

Starter scope: labelled Set A only. CPU baseline feasible. Substantial sequence preprocessing.



13 Air-quality forecast uncertainty

Question	**Are forecast intervals reliable during pollution spikes and missing data?**
Input	UCI Beijing Multi-Site hourly pollutants, weather, station IDs and NA values.
Task	Forecast PM2.5 using quantile boosting. Compare an LSTM or point model and a missingness ablation.
Deliver	24-hour forecasts, 80% intervals, coverage/width by station and spike-error analysis.

Data: Beijing Multi-Site GitHub: lightgbm-org/LightGBM Paper: Zhang (2017)

Starter scope: 2 stations, then expand. CPU feasible for boosting. Causal preprocessing required.



14 Text leakage and shortcut features

Question	**Is high topic accuracy coming from content or metadata shortcuts?**
Input	20 Newsgroups documents with headers, footers and quotations available for controlled removal.
Task	Compare TF-IDF + linear classification with a CNN. Repeat after stripping metadata and auditing duplicates.
Deliver	A leakage audit, macro F1 before/after cleaning and explanations of influential features.

Data: scikit-learn 20 Newsgroups GitHub: scikit-learn Paper: Lang (1995)

Starter scope: 4 related topics, then all 20. CPU baseline feasible. GPU helpful for CNN.



15 Sentiment and long-range context

Question	**Which model handles negation and long reviews under the same budget?**
Input	IMDb labelled raw reviews and official train/test split.
Task	Compare TF-IDF logistic regression, a text CNN and an LSTM at two truncation lengths.
Deliver	Sentiment probabilities, F1, length/negation error slices and a quality/runtime comparison.

Data: Stanford IMDb GitHub: keras-team/keras-io Paper: Maas (2011)

Starter scope: 10,000 training reviews, 2 sequence lengths. GPU helpful for CNN/LSTM.



16 Active learning with a label budget

Question	**Which articles should be labelled next to improve a news classifier?**
Input	AG News texts and labels. Hide pool labels until the simulated query requests them.
Task	Compare random, uncertainty and diversity selection with the same classifier and annotation budgets.
Deliver	F1 versus labels at 200, 500 and 1,000 examples, query logs and variation across seeds.

Data: AG News GitHub: modAL-python/modAL Paper: Zhang (2015)

Starter scope: a 10,000-document pool and 3 seeds. CPU feasible with a linear model.



17 Explanations for hate-speech models

Question	**Do a model's highlighted words support its prediction?**
Input	HateXplain posts with hate/offensive/normal labels, target communities and human rationales.
Task	Compare TF-IDF and CNN/LSTM models. Test rationale overlap and deletion-based explanations.
Deliver	Macro F1, rationale metrics, sufficiency/comprehensiveness and target-community error slices.

Data: HateXplain GitHub: hate-alert/HateXplain Paper: Mathew (2021)

Starter scope: official splits and one explanation method. GPU helpful. Annotated social text.



18 Intent recognition with abstention

Question	**When should an assistant reject an unsupported request?**
Input	CLINC150 in-scope utterances, 150 intent labels and out-of-scope validation/test examples.
Task	Compare classifier confidence with autoencoder reconstruction on fixed text features for rejection.
Deliver	Intent predictions or abstentions, OOS detection scores and coverage-versus-error curves.

Data: CLINC150 full JSON GitHub: clinc/oos-eval Paper: Larson (2019)

Starter scope: full data_full.json and compact TF-IDF/SVD features. CPU feasible.



19 Multiple emotions in one comment

Question	**Can a model predict several emotions without ignoring rare labels?**
Input	GoEmotions comments with multiple emotion IDs and the official filtered splits.
Task	Compare one-vs-rest boosting with a sigmoid CNN/LSTM. Tune label thresholds on validation data.
Deliver	Label-probability vectors, macro/micro F1, rare-label results and a threshold ablation.

Data: GoEmotions filtered data GitHub: Google / GoEmotions Paper: Demszky (2020)

Starter scope: official filtered release, 27 emotions plus Neutral. CPU baseline feasible.



20 Multilingual intent under unequal labels

Question	**How does one model serve languages with different annotation budgets?**
Input	MASSIVE 1.0 utterances, intent labels, locale, shared IDs and official partitions.
Task	Select 3 languages. Compare character CNN/LSTM with TF-IDF + XGBoost under balanced and unequal budgets.
Deliver	Per-language macro F1, budget curves, translation-ID leakage checks and error examples.

Data: MASSIVE 1.0 GitHub: alexa/massive Paper: FitzGerald (2022)

Starter scope: 3 verified locales and intent prediction only. GPU helpful for CNN/LSTM.



21 Hierarchical image classification

Question	**Does learning broad categories help distinguish similar objects?**
Input	CIFAR-100 images with 100 fine labels and 20 superclass labels.
Task	Compare a fine-label CNN with a CNN using an auxiliary superclass loss. Add frozen features + XGBoost.
Deliver	Fine/coarse accuracy, hierarchy-consistency errors and results as the auxiliary loss weight changes.

Data: CIFAR-100 GitHub: pytorch/vision Paper: Krizhevsky (2009)

Starter scope: 10,000 training images covering all classes. GPU helpful.

22 Learning with noisy image labels

Question	**How much label noise can a classifier tolerate?**
Input	Fashion-MNIST images. Inject known label flips into training data while retaining clean held-out labels.
Task	Compare a CNN and XGBoost at 0%, 10% and 30% label noise. Test label smoothing with matched budgets.
Deliver	Clean-test macro F1, noise-response curves, per-class errors and variation across 3 corruption seeds.

Data: Fashion-MNIST GitHub: fashion-mnist Paper: Xiao (2017)

Starter scope: 10,000 training images and symmetric label flips. CPU feasible for a small CNN.



23 Learning from unlabelled images

Question	**Can unlabelled images improve a classifier with the same label budget?**
Input	STL-10 labelled images, official test images and its separate unlabelled image pool.
Task	Compare supervised training with autoencoder pretraining or pseudo-labelling. Keep the labelled subset fixed.
Deliver	Macro F1 versus labels, reconstruction or pseudo-label diagnostics and a confidence-threshold ablation.

Data: STL-10 GitHub: pytorch/vision Paper: Coates (2011)

Starter scope: 1,000 labelled and 5,000 unlabelled training images. GPU helpful.

24 Compact CNNs through distillation

Question	**Can a compact student retain the accuracy of a larger teacher?**
Input	CIFAR-10 training images and a teacher CNN trained only on the same permitted training data.
Task	Train a small CNN with hard labels alone or with teacher soft targets. Vary distillation temperature.
Deliver	Accuracy, calibration, parameter counts and CPU latency for the teacher and both student settings.

Data: CIFAR-10 GitHub: PyTorch tutorials Paper: Hinton (2015)

Starter scope: 10,000 training images, one teacher and one student. GPU helpful for teacher training.



25 Rare traffic signs and class imbalance

Question	**Can a model improve rare-class recall without losing common-class accuracy?**
Input	GTSRB traffic-sign images, class IDs and training track IDs grouping views of the same physical sign.
Task	Compare HOG + XGBoost and a CNN. Test class-weighted loss or balanced sampling with fixed validation tracks.
Deliver	Macro F1, rare-class recall, class-frequency plots and a confusion matrix for similar signs.

Data: GTSRB GitHub: GTSRB loader Paper: Stallkamp (2012)

Starter scope: 10 classes spanning several frequencies. Split validation by track. GPU helpful.

26 Matching unseen handwritten characters

Question	**Can a model match two examples of a character absent from training?**
Input	Omniglot character images with alphabet and character IDs. Form same-character and different-character pairs.
Task	Compare a Siamese CNN with pixel-distance and fixed-feature baselines. Hold out entire character classes.
Deliver	Pair scores, verification AUROC, balanced accuracy and errors on novel characters and alphabets.

Data: Omniglot GitHub: brendenlake/omniglot Paper: Lake (2015)

Starter scope: 300 training characters and balanced image pairs. Small CNN feasible on CPU.



27 Adversarial image perturbations

Question	**How easily can bounded pixel changes fool a classifier?**
Input	MNIST digit images, clean test labels and adversarial variants generated with declared pixel bounds.
Task	Train a CNN and compare FGSM with multi-step PGD. Test adversarial training using the same attack budget.
Deliver	Clean/robust accuracy, curves versus perturbation size, attack examples and measured runtime.

Data: MNIST loader GitHub: bethgelab/foolbox Paper: Rauber (2017)

Starter scope: 10,000 training and 1,000 test images. Small CNN feasible on CPU.



28 Image denoising and detail preservation

Question	**Can a denoiser remove noise without erasing fine image details?**
Input	BSDS500 natural images with synthetic Gaussian noise. Use the original clean images as targets.
Task	Compare a convolutional denoiser with non-local means. Train at one noise level and test several levels.
Deliver	Denoised images, PSNR/SSIM, detail-error crops and performance across noise levels.

Data: BSDS500 GitHub: scikit-image Paper: Arbelaez (2011)

Starter scope: grayscale 128 × 128 patches. Split source images before creating patches. GPU helpful.

29 Texture features and spatial pooling

Question	**Which representation distinguishes textures when object shape offers little help?**
Input	Describable Textures Dataset images, 47 texture labels and official train/validation/test split files.
Task	Compare LBP histograms + XGBoost with frozen CNN features. Test global versus pooled local descriptors.
Deliver	Macro F1, texture confusion examples and a feature/pooling ablation with extraction times.

Data: DTD GitHub: DTD loader Paper: Cimpoi (2014)

Starter scope: one official split and cached CNN features. CPU feasible after feature extraction.



30 Continual learning and forgotten classes

Question	**Can a classifier learn new classes while retaining earlier ones?**
Input	Kuzushiji-MNIST images and 10 labels. Reveal training classes in five phases of two classes each.
Task	Compare sequential CNN training with a fixed replay buffer. Include joint training as an upper reference.
Deliver	An accuracy matrix after each phase, forgetting scores and results versus replay-memory size.

Data: Kuzushiji-MNIST GitHub: rois-codh/kmnist Paper: Clanuwat (2018)

Starter scope: five 2-class phases, one small CNN and 3 class orders. CPU feasible.



31 Spoken keywords for unseen speakers

Question	**Which audio representation recognizes short commands for new speakers?**
Input	Speech Commands v0.02 one-second WAV clips, word labels and speaker identifiers in filenames.
Task	Compare MFCC summaries + XGBoost with a spectrogram CNN. Test training-only background-noise augmentation.
Deliver	Keyword probabilities, macro F1, errors by speaker and clean/noisy performance at fixed noise levels.

Data: Speech Commands GitHub: TensorFlow examples Paper: Warden (2018)

Starter scope: 10 command words and official partitions. GPU helpful for the spectrogram CNN.



32 Sound events and recording bias

Question	**Does a sound classifier recognize events beyond familiar recordings?**
Input	ESC-50 audio clips with sound labels, five predefined folds and original recording identifiers.
Task	Compare MFCC summaries + boosting with a log-mel CNN. Preserve official folds and test time masking.
Deliver	Macro F1 across folds, class confusions, audio examples and an augmentation ablation.

Data: ESC-50 / ESC-10 GitHub: karolpiczak/ESC-50 Paper: Piczak (2015)

Starter scope: the 10-class ESC-10 subset and two held-out folds. GPU helpful for CNN training.

33 ECG classification with fewer sensor leads

Question	**Which ECG leads are necessary for predicting broad diagnostic labels?**
Input	PTB-XL v1.0.3 100 Hz waveforms, diagnostic superclass labels, patient IDs and recommended folds.
Task	Compare signal summaries + XGBoost with a 1D CNN. Repeat with 12 leads and a fixed smaller lead subset.
Deliver	Per-label AUROC/AUPRC, macro F1, a lead ablation and errors on patients absent from training.

Data: PTB-XL v1.0.3 GitHub: PTB-XL benchmarks Paper: Wagner (2020)

Starter scope: 5 diagnostic superclasses and 100 Hz records. GPU helpful. Research classification task.



34 Sleep stages and temporal context

Question	**Does nearby EEG context improve sleep-stage classification?**
Input	Sleep-EDF Expanded sleep-cassette EEG recordings and expert hypnograms, grouped by participant.
Task	Compare spectral features + XGBoost with a 1D CNN. Add previous-epoch context without future epochs.
Deliver	Stage predictions, macro F1, transition errors and results for one epoch versus past-context windows.

Data: Sleep-EDF Expanded GitHub: mne-tools/mne-python Paper: Kemp (2000)

Starter scope: one EEG channel from 20 participants. Split by person. GPU helpful. EDF preprocessing.



35 Sensor augmentation and label preservation

Question	**Which augmentations improve a sensor classifier without changing its label?**
Input	UCR FordA sensor sequences and binary labels with the archive's fixed training/test partition.
Task	Compare a 1D CNN and summary-feature XGBoost. Test jitter and scaling separately with matched updates.
Deliver	Macro F1 across seeds, augmentation-strength curves and signal examples that expose label distortion.

Data: UCR FordA GitHub: timeseriesAI/tsai Paper: Dau (2019)

Starter scope: FordA and two augmentation types. Small CNN feasible on CPU. Keep test signals intact.



36 Appliance power from household totals

Question	**Can household power reveal when one appliance runs and how much energy it uses?**
Input	UK-DALE 6-second household mains and appliance submeter readings with synchronized timestamps.
Task	Compare a sequence CNN with XGBoost on mains-window features. Hold out later time blocks.
Deliver	Appliance power traces, MAE, on/off F1 and daily energy error, including missed activations.

Data: UK-DALE GitHub: nilmtk/nilmtk Paper: Kelly (2015)

Starter scope: one appliance in one house over four weeks. CPU baseline feasible. Large raw download.



37 Reconstructing long sensor gaps

Question	**Which method restores contiguous missing measurements most accurately?**
Input	Observed ETTh1 sensor values with reproducible artificial gaps of several lengths and a saved gap mask.
Task	Compare linear interpolation with a masked sequence autoencoder. Evaluate offline reconstruction of gaps.
Deliver	MAE/RMSE on hidden values, errors by gap length, reconstruction plots and an edge-gap analysis.

Data: ETTh1 CSV GitHub: timeseriesAI/tsai Paper: Zhou (2021)

Starter scope: one sensor, 6/24/48-hour gaps and 3 masks. GPU helpful for the autoencoder.



38 Traffic forecasting under concept drift

Question	**Does periodic retraining keep traffic forecasts reliable as patterns change?**
Input	UCI Metro Interstate Traffic Volume timestamps, hourly traffic and calendar/weather records.
Task	Compare a frozen LightGBM forecast with monthly rolling retraining. Include a seasonal-naive baseline.
Deliver	One-hour forecasts, monthly MAE, a drift/retraining timeline and the cost of model updates.

Data: Metro traffic volume GitHub: LightGBM Paper: Gama (2014)

Starter scope: hourly data, 24/168-hour lags and monthly evaluation. CPU feasible.



39 Variable-length sequence classification

Question	**Does padding hide useful information in sequences of different lengths?**
Input	UEA JapaneseVowels utterances with 12 acoustic coefficient channels, speaker labels and variable lengths.
Task	Compare summary-feature XGBoost with a masked LSTM. Test masked versus unmasked pooling after padding.
Deliver	Speaker-class macro F1, errors by sequence length and a padding/masking ablation.

Data: JapaneseVowels GitHub: timeseriesAI/tsai Paper: Bagnall (2018)

Starter scope: the full small archive dataset. CPU feasible. This task predicts speaker identity.

40 Early decisions from partial sequences

Question	**How early can a model classify a sequence without a large accuracy loss?**
Input	UCR Wafer sensor sequences and normal/abnormal labels. Reveal only a prefix at each decision point.
Task	Compare XGBoost prefix summaries with a 1D CNN at 25%, 50%, 75% and 100% of the sequence.
Deliver	F1 versus observed length, accuracy/earliness trade-offs and examples of delayed or premature decisions.

Data: UCR Wafer GitHub: timeseriesAI/tsai Paper: Dau (2019)

Starter scope: Wafer and four fixed prefix lengths. CPU feasible. Fit preprocessing from prefixes only.



41 Spam detection with unequal error costs

Question	**Which decision threshold protects useful messages while catching spam?**
Input	UCI SMS Spam Collection messages and ham/spam labels. Define false-positive and false-negative costs.
Task	Compare TF-IDF logistic regression with a character CNN. Tune thresholds for several declared cost ratios.
Deliver	Precision/recall curves, confusion matrices, cost per 1,000 messages and representative false positives.

Data: SMS Spam Collection GitHub: scikit-learn Paper: Almeida (2011)

Starter scope: the full small dataset and three cost ratios. CPU baseline feasible.



42 Paraphrases beyond shared words

Question	**Can a model distinguish equivalent sentences from pairs with similar vocabulary?**
Input	GLUE MRPC sentence pairs and binary paraphrase labels from its labelled train/validation partitions.
Task	Compare word-overlap + XGBoost with a Siamese CNN/LSTM. Analyse high-overlap negative pairs.
Deliver	Pair probabilities, F1, overlap-based error slices and examples where meaning changes despite shared words.

Data: GLUE MRPC GitHub: TensorFlow Datasets Paper: Dolan (2005)

Starter scope: labelled MRPC only. Use training data for tuning and reserve validation for evaluation.



43 Semantic similarity as a continuous score

Question	**Does a model's similarity score follow human judgements beyond exact word overlap?**
Input	STS Benchmark sentence pairs and human similarity scores from labelled training/development data.
Task	Compare TF-IDF cosine similarity with XGBoost pair regression and a shared LSTM sentence encoder.
Deliver	Pearson/Spearman correlations, MAE, a prediction scatterplot and low-overlap/high-similarity errors.

Data: GLUE STS-B GitHub: TensorFlow Datasets Paper: Cer (2017)

Starter scope: labelled STS-B only. Reserve development for final evaluation. CPU baseline feasible.



44 Question types and word order

Question	**Does word order help identify the kind of answer a question requests?**
Input	TREC question-classification text with coarse/fine answer-type labels and fixed training/test files.
Task	Compare unigram and bigram TF-IDF models with a text CNN. Test shuffled-token inputs as an ablation.
Deliver	Macro F1, class confusions, order-sensitive examples and performance before/after token shuffling.

Data: TREC questions GitHub: scikit-learn Paper: Li & Roth (2002)

Starter scope: six coarse classes. CPU baseline feasible. Extend to fine classes after the core result.



45 Named entities in noisy social text

Question	**Can a model locate unfamiliar names despite spelling variation and informal writing?**
Input	WNUT 2017 token sequences with BIO entity labels and official train/development/test files.
Task	Compare a feature-based sequence tagger with a BiLSTM. Test the benefit of character-level inputs.
Deliver	Entity spans, exact-span F1, rare/unseen-name errors and a word-only versus character-input ablation.

Data: WNUT 2017 GitHub: emerging_entities_17 Paper: Derczynski (2017)

Starter scope: the full small release and entity tagging only. GPU helpful for BiLSTM training.



46 Grammar tagging for unfamiliar words

Question	**Can subword features identify grammatical roles for words absent from training?**
Input	Universal Dependencies English EWT sentences, token forms and universal part-of-speech tags.
Task	Compare context-feature logistic regression with a BiLSTM. Test word-only versus character-aware inputs.
Deliver	Token tags, accuracy/macro F1, unknown-word results and confusions between similar grammatical roles.

Data: UD English EWT GitHub: UD_English-EWT Paper: Nivre (2020)

Starter scope: one pinned UD English EWT release and UPOS tags only. CPU baseline feasible.



47 Sentence inference and shortcut learning

Question	**Does a model compare both sentences or guess from the hypothesis alone?**
Input	SNLI premise/hypothesis pairs with entailment, contradiction and neutral labels and official splits.
Task	Compare a hypothesis-only baseline with a paired CNN/LSTM encoder. Analyse negation and lexical-overlap slices.
Deliver	Three-class probabilities, macro F1, shortcut gaps and examples where the premise changes the decision.

Data: Stanford SNLI GitHub: TensorFlow Datasets Paper: Bowman (2015)

Starter scope: 30,000 training pairs and all three labels. GPU helpful for paired encoders.

48 Retrieving evidence for scientific claims

Question	**Can a retriever find the abstracts that contain evidence for a claim?**
Input	SciFact claims, scientific abstracts, evidence document IDs and labelled claim partitions.
Task	Compare TF-IDF retrieval with a trainable pair ranker. Re-rank retrieved candidates using only training claims.
Deliver	Ranked abstracts, Recall@5/10, MRR and missed-evidence examples with a candidate-retrieval audit.

Data: SciFact GitHub: allenai/scifact Paper: Wadden (2020)

Starter scope: labelled train/dev claims and abstract retrieval only. CPU feasible for lexical models.



49 Topic discovery without category labels

Question	**Do unsupervised document groups recover meaningful news themes?**
Input	BBC news articles from the UCD corpus. Hide category labels during preprocessing and model selection.
Task	Compare NMF topic representations with TF-IDF K-means and LDA. Inspect topic stability across seeds.
Deliver	Topic words, representative articles, cluster stability and ARI/NMI against labels revealed only at evaluation.

Data: BBC corpus mirror GitHub: scikit-learn Paper: Greene (2006)

Starter scope: the small BBC corpus. CPU feasible. Reveal category labels only for the final audit.



50 Review ratings as an ordered target

Question	**Does modelling rating order reduce large mistakes between one and five stars?**
Input	Amazon Reviews 2023 text, 1-5 star ratings, product/user IDs and timestamps from one category.
Task	Compare TF-IDF/SVD + CatBoost regression with an LSTM classifier. Test an ordinal loss on the same LSTM.
Deliver	MAE, quadratic weighted kappa, rating-confusion plots and errors by star frequency and product group.

Data: Amazon Reviews 2023 GitHub: AmazonReviews2023 Paper: Hou (2024)

Starter scope: 20,000 All_Beauty reviews with a saved sampling manifest. GPU helpful for LSTM training.

Notebook submission

Minimum	One executed .ipynb reproduces the core results through Restart & Run All.
Data pipeline	Include acquisition instructions, raw-data audit, split and all preprocessing code.
Experiment	Include a baseline, an advanced method, a controlled comparison and error analysis.
Supporting files	If needed, submit all scripts/configs and environment details with the notebook.

Prefer one notebook. Raw data may live in separate folders. Every required transformation must be supplied.

Suggested assessment

Evidence	Suggested weight
Data audit and reproducible pipeline	20%
Baseline and controlled comparison	20%
Valid splits, tuning and evaluation	25%
Error analysis and interpretation	20%
Documentation and source attribution	15%
Total	100%

A well-supported negative result earns credit. Include one small hand calculation.

Proposed rubric. Agree on the dataset subset and compute budget before model tuning.